

Decision Tree Model and Business Interpretation Based on a Business Plan

FitLife Studio Analytics: Membership Renewal Risk Prediction

Prepared for: Assignment 4 - Decision Tree Classification and Business
Interpretation

Course: MBAI 5310G - AI Programming

Prepared by: Vettivelu Prabaseelan

Model Used: Decision Tree Classifier

Target Variable: At_Risk_Not_Renewing

Positive Class: Yes

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1. Executive Summary

This report presents the Assignment 4 Decision Tree classification project for FitLife Studio Analytics. The project uses a structured machine learning workflow to predict whether a fitness studio member is at risk of not renewing their membership. The report connects technical model results to business decision-making, customer retention planning, responsible AI use, and practical management recommendations.

The dataset originally contained 416 rows and 21 columns. After removing 6 duplicate rows, the final cleaned dataset contained 410 rows and 21 columns. The target variable is `At_Risk_Not_Renewing`, where Yes indicates that the member is at risk of not renewing.

The Decision Tree model achieved a testing accuracy of 74.39%. For the positive class, Yes, the model achieved precision of 72.00%, recall of 56.25%, and F1-score of 63.16%. The model also showed a training-testing accuracy difference of 3.66%, which suggests that the selected tree depth provides a reasonable balance between interpretability and generalization.

From a business perspective, the model is useful as a decision-support tool. It helps FitLife identify members who may require proactive retention support, such as service follow-up, class recommendations, renewal reminders, billing assistance, or personalized engagement. However, because the recall for the at-risk class is moderate, the model should not be used as the only basis for retention decisions.

2. Introduction and Assignment Objective

The objective of this assignment is to build and interpret a Decision Tree classification model based on a business plan. The assignment requires both technical implementation and business interpretation. Therefore, the report covers the full workflow from business understanding and dataset preparation to model training, evaluation, feature importance, limitations, and responsible use.

The problem is a supervised binary classification task because the target variable has two possible outcomes: Yes and No. The model uses member demographics, membership information, engagement behaviour, satisfaction indicators, payment history, and communication data to predict renewal-risk status.

This report is organized to present the business problem, data preparation, model development, evaluation results, business interpretation, limitations, and responsible AI considerations in a clear and professional structure.

3. Business Context and Machine Learning Decision

FitLife Studio is a fitness subscription business that earns an important share of revenue from recurring membership renewals. Since members can switch to other gyms, lower-cost fitness options, or home workout alternatives, early identification of non-renewal risk is important for retention management.

The business problem is membership churn. Members may become less engaged over time by visiting less often, attending fewer classes, using the app less frequently, reporting lower

satisfaction, experiencing payment issues, or failing to respond to renewal reminders. These behaviours may signal that the member is at risk of not renewing.

The machine learning decision supported by the model is: Should FitLife proactively contact this member with a retention action before the renewal period ends?

Table 1: Business challenge and possible retention action

Business Challenge	Why It Matters	Possible Retention Action
Low member engagement	Members may feel they are not receiving enough value from the membership.	Send personalized class suggestions, goal check-ins, or app engagement reminders.
Low satisfaction or complaints	Dissatisfied members are more likely to leave.	Prioritize service recovery conversations and manager follow-up.
Payment issues	Billing friction may increase cancellation risk.	Offer billing support or flexible payment assistance.
Monthly contract type	Monthly members can leave more easily than members on longer contracts.	Provide early renewal incentives or explain longer-term plan benefits.
Long distance from studio	Convenience can affect renewal decisions.	Recommend online classes or a more convenient branch option.

4. Dataset Overview

The dataset used for this project is `fitlife_membership_renewal_risk_dataset.xlsx`. Each row represents one FitLife Studio member. The dataset includes demographic variables, membership details, engagement variables, satisfaction and complaint indicators, payment variables, renewal communication, and the target renewal-risk outcome.

Table 2: Dataset and project summary

Item	Value
Original dataset shape	416 rows x 21 columns
Duplicate rows before cleaning	6
Duplicate rows after cleaning	0
Final cleaned dataset shape	410 rows x 21 columns
Target variable	At Risk Not Renewing
Positive class	Yes = member is at risk of not renewing
Problem type	Supervised binary classification
Model used	Decision Tree Classifier

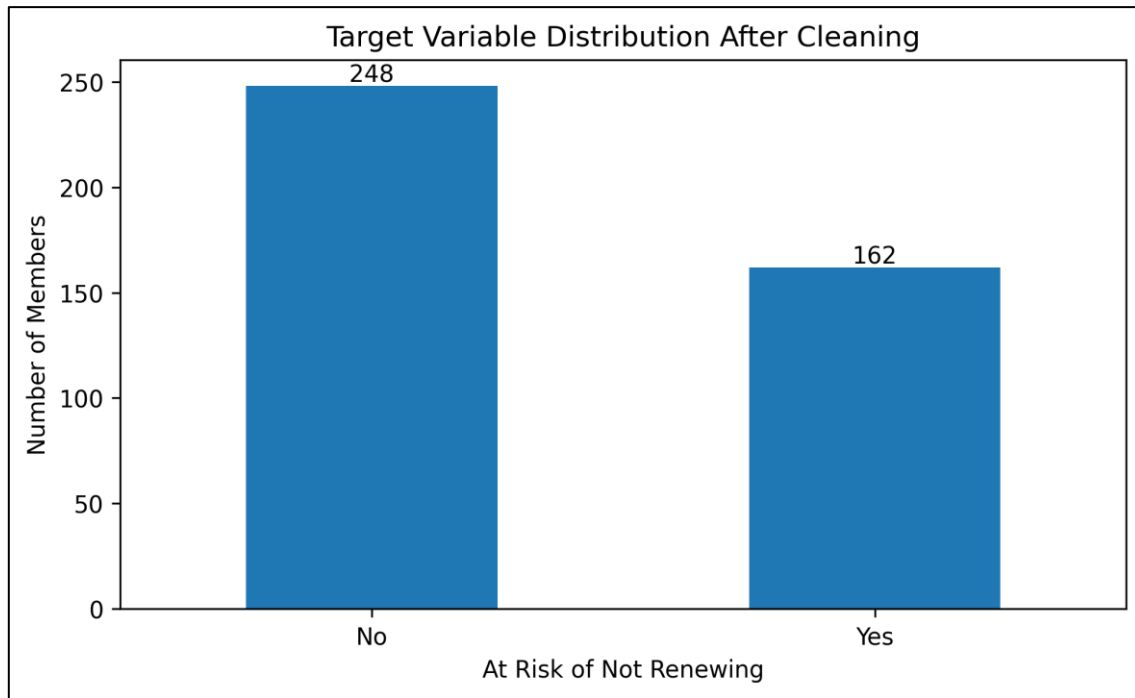


Figure 1: Target variable distribution after cleaning

Figure 1 shows that the cleaned dataset contains 248 members in the No class and 162 members in the Yes class. The No class is larger than the Yes class, so precision, recall, and confusion matrix values should be interpreted together with accuracy.

5. Target Variable and Input Features

The target variable is `At_Risk_Not_Renewing`. The value Yes means that the member is considered at risk of not renewing, while No means that the member is not currently considered high risk. The model uses the remaining business variables as input features after removing `Member_ID`.

Table 3: Dataset columns and business meaning

Column	Type	Business Meaning
Member_ID	Identifier	Identifier only; removed before model training.
Age	Numeric	Member age in years.
Gender	Categorical	Self-reported gender category.
Membership_Type	Categorical	Membership tier purchased by the customer.
Monthly_Fee	Numeric	Monthly membership fee.
Months_As_Member	Numeric	Length of customer relationship.
Contract_Type	Categorical	Membership contract structure.
Avg_Weekly_Visits	Numeric	Average number of studio visits per week.
Classes_Per_Month	Numeric	Number of fitness classes attended in the last month.
Personal_Training	Categorical	Whether the member uses paid personal training.

Column	Type	Business Meaning
App_Login_Days_Last_30	Numeric	Number of app login days in the last 30 days.
Payment_Issues_Last_6_Months	Numeric	Number of payment problems in the last six months.
Satisfaction_Score	Numeric	Member satisfaction score from survey data.
Complaint_Count_Last_3_Months	Numeric	Number of complaints in the last three months.
Distance_From_Studio_km	Numeric	Estimated distance from home area to nearest studio.
Last_Renewal_Discount_Used	Categorical	Whether a discount was used at the previous renewal.
Renewal_Reminder_Sent	Categorical	Whether a renewal reminder was sent.
Corporate_Plan	Categorical	Whether the member belongs to a corporate partner plan.
Season	Categorical	Season when the renewal decision is observed.
Referral_Source	Categorical	Main acquisition channel for the member.
At_Risk_Not_Renewing	Target	Target variable; Yes or No renewal-risk outcome.

Table 4: Numerical and categorical feature groups

Feature Type	Number of Features	Columns
Numerical features	10	Age, Monthly_Fee, Months_As_Member, Avg_Weekly_Visits, Classes_Per_Month, App_Login_Days_Last_30, Payment_Issues_Last_6_Months, Satisfaction_Score, Complaint_Count_Last_3_Months, Distance_From_Studio_km
Categorical features	9	Gender, Membership_Type, Contract_Type, Personal_Training, Last_Renewal_Discount_Used, Renewal_Reminder_Sent, Corporate_Plan, Season, Referral_Source

6. Data Preparation and Data Quality Review

Data preparation was completed before model training to improve data quality and make the dataset suitable for machine learning. The inspection included checking the dataset shape, column names, data types, duplicate rows, missing values, target distribution, numerical features, and categorical features.

Table 5: Target variable distribution before and after cleaning

Stage	No	Yes	Total
Before duplicate removal	253	163	416
After duplicate removal	248	162	410

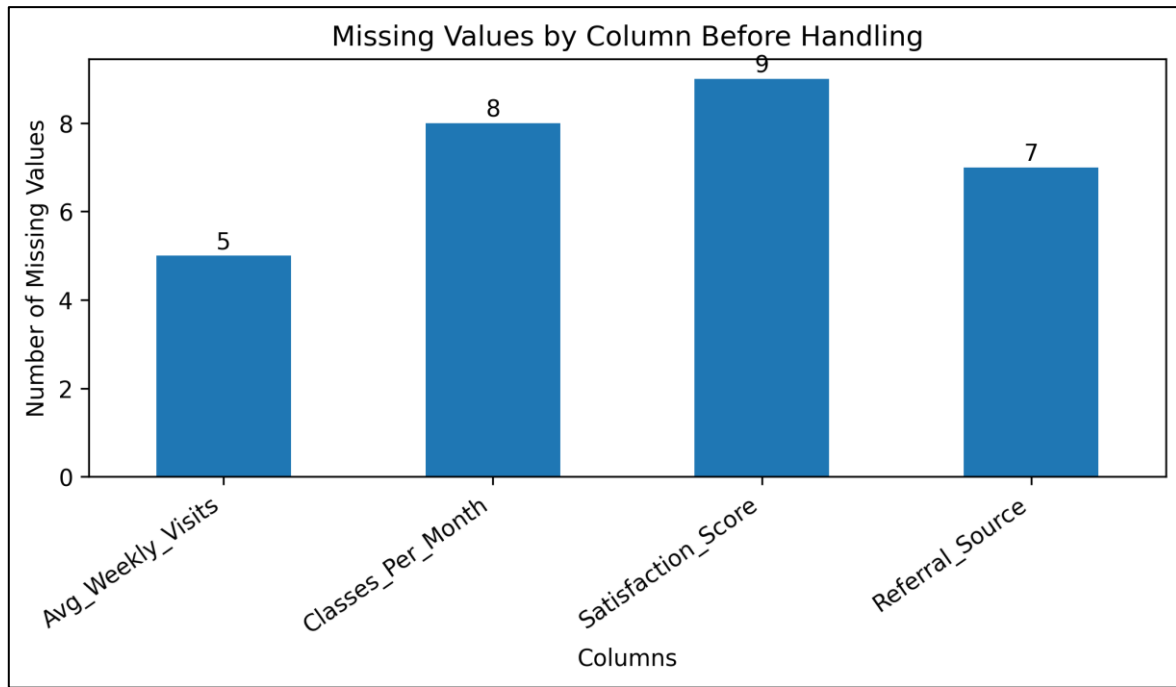


Figure 2: Missing values by column before handling

Figure 2 shows that missing values appeared in three numerical variables and one categorical variable. The target variable had no missing values and was not filled because it is the outcome the model is designed to learn.

Table 6: Missing value inspection and handling plan

Column	Data Type	Missing Values	Planned Method	Handling Location
Avg_Weekly_Visits	Numerical	5	Median	Inside preprocessing pipeline
Classes Per Month	Numerical	8	Median	Inside preprocessing pipeline
Satisfaction_Score	Numerical	9	Median	Inside preprocessing pipeline
Referral_Source	Categorical	7	Most frequent value	Inside preprocessing pipeline

Missing values were not filled directly in the full dataset before the train-test split. Instead, missing values were reviewed after duplicate removal and handled later inside the preprocessing pipeline. Numerical missing values were handled using median imputation, while categorical missing values were handled using the most frequent value. This approach helps avoid data leakage because the preprocessing transformer is fitted on the training data and then applied to the testing data.

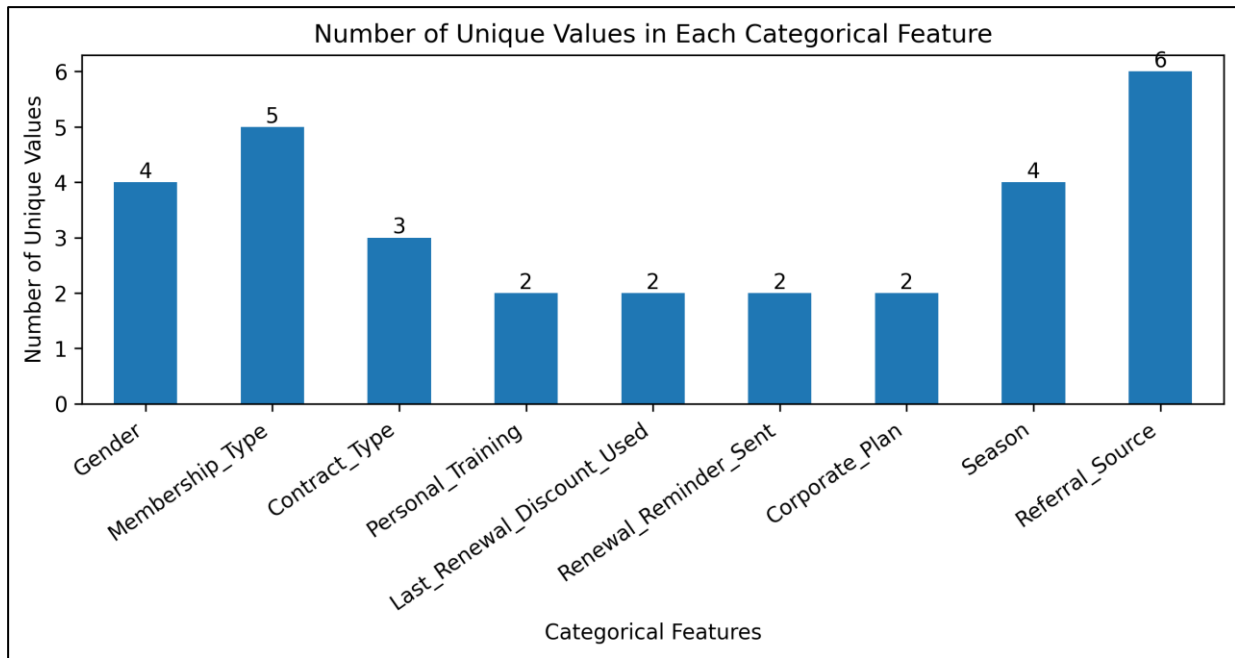


Figure 3: Number of unique values in each categorical feature

7. Feature/Target Separation and Data Splitting

After cleaning, the dataset was separated into the feature matrix X and the target vector y. The target variable At_Risk_Not_Renewing was separated from the inputs, and Member_ID was removed because it is an identifier rather than a business behaviour variable.

Table 7: Feature matrix and target vector

Object	Shape	Meaning
X	410 rows x 19 feature columns	Input variables used by the model
y	410 values	Target variable: At Risk Not Renewing

Table 8: Train-test split summary

Dataset	Shape
X_train	(328, 19)
X_test	(82, 19)
y_train	(328,)
y_test	(82,)

Table 9: Target class distribution in training and testing sets

Target Class	Training Set	Testing Set
No	198	50
Yes	130	32

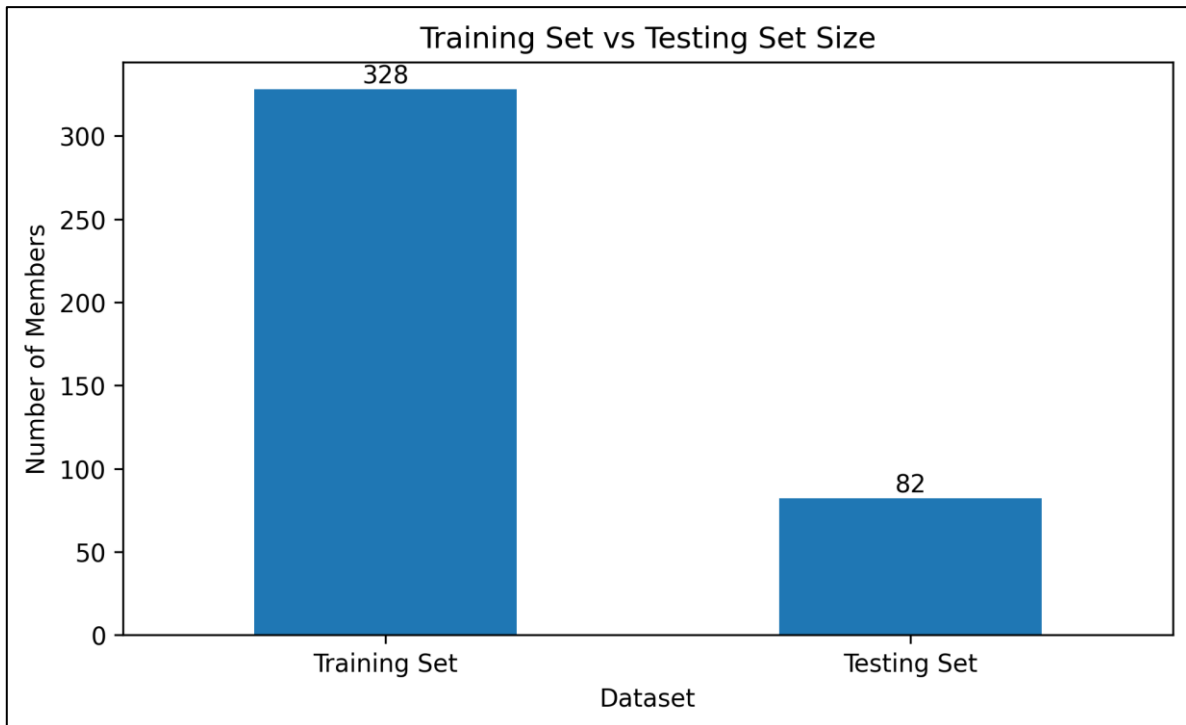


Figure 4: Training set vs testing set size

The dataset was split using an 80/20 split with `random_state=42` and `stratify=y`. Stratification helps maintain a similar target class distribution in the training and testing datasets.

8. Preprocessing and Decision Tree Modelling

Machine learning models require numerical input. Therefore, preprocessing was applied to both numerical and categorical variables. Numerical missing values were handled using median imputation, and numerical variables were not scaled because Decision Tree models do not require feature scaling. Categorical missing values were handled using the most frequent value, and categorical variables were converted into numerical columns using one-hot encoding model.

Table 10: Preprocessing approach

Feature Group	Columns	Transformation
Numerical features	Age, Monthly_Fee, Months_As_Member, Avg_Weekly_Visits, Classes_Per_Month, App_Login_Days_Last_30, Payment_Issues_Last_6_Months, Satisfaction_Score, Complaint_Count_Last_3_Months, Distance_From_Studio_km	SimpleImputer(strategy="median"); no scaling
Categorical features	Gender, Membership_Type, Contract_Type, Personal_Training, Last_Renewal_Discount_Used, Renewal_Reminder_Sent,	SimpleImputer(strategy="most_frequent") and OneHotEncoder(handle_unknown="ignore")

	Corporate_Plan, Season, Referral_Source	
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Table 11: Decision Tree model configuration

Model Item	Value
Algorithm	DecisionTreeClassifier
Maximum depth	4
Random state	42
Training input shape after preprocessing	(328, 40)
Testing input shape after preprocessing	(82, 40)
Number of processed features	40

The max_depth=4 setting was used to keep the tree reasonably simple and interpretable. This is appropriate for an assignment focused on business interpretation because a very deep tree can become harder to explain and more likely to overfit the training data.

9. Model Evaluation Results

The trained Decision Tree model was evaluated on the testing data. Predictions were compared with actual values using accuracy, precision, recall, F1-score, a classification report, and a confusion matrix.

Table 12: Sample actual and predicted values

Actual	Predicted
No	No
Yes	Yes
Yes	No
No	No
Yes	No
Yes	No
No	Yes
Yes	No
No	No
Yes	No

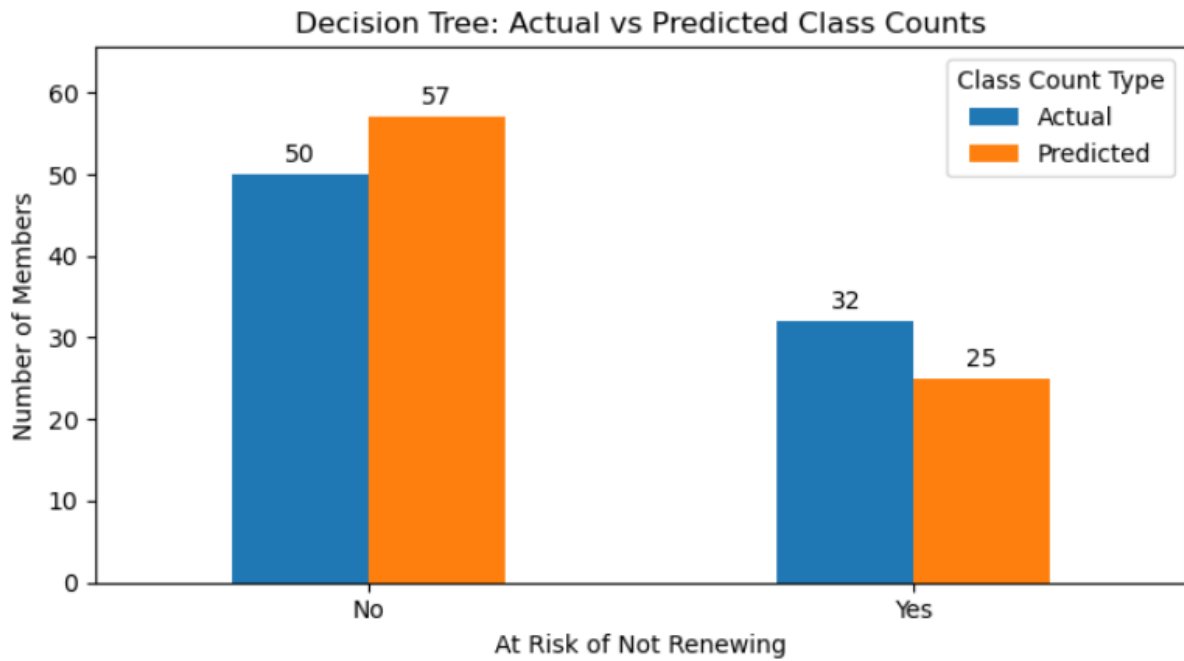


Figure 5: Actual vs predicted class counts

Table 13: Decision Tree evaluation metrics

Metric	Score	Percentage
Accuracy	0.7439	74.39%
Precision for Yes	0.7200	72.00%
Recall for Yes	0.5625	56.25%
F1-score for Yes	0.6316	63.16%

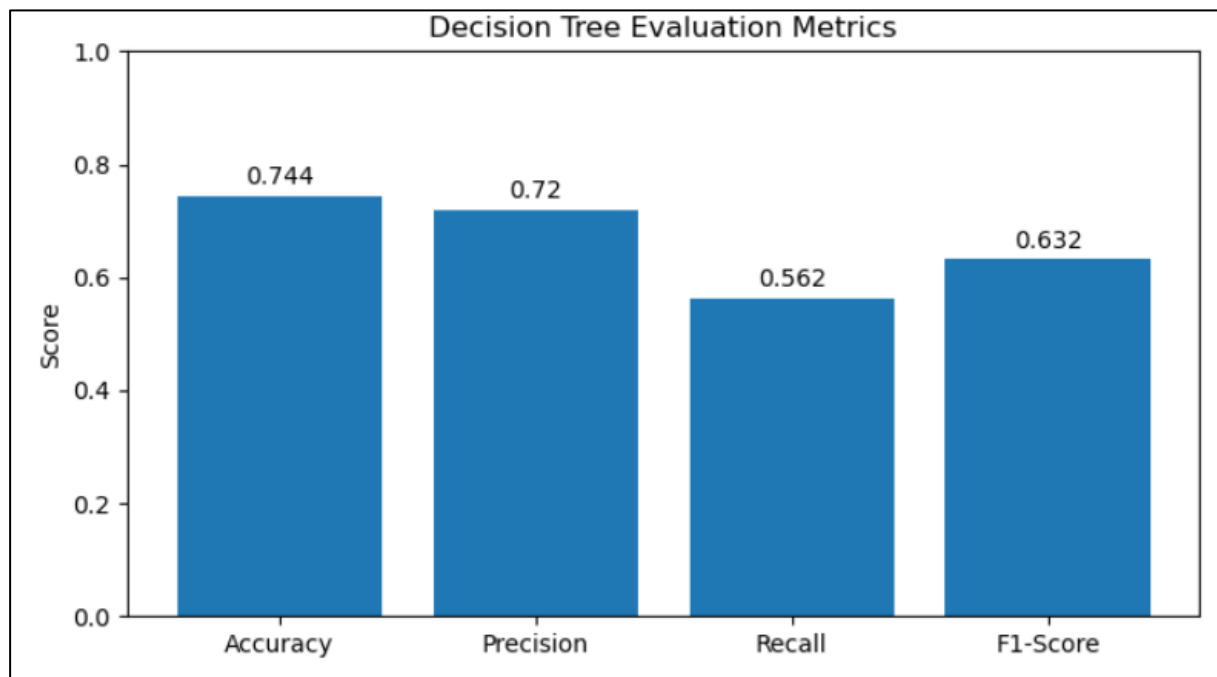


Figure 6: Decision Tree evaluation metrics

Table 14: Classification report

Class/Metric	Precision	Recall	F1-score	Support
No	0.7544	0.8600	0.8037	50
Yes	0.7200	0.5625	0.6316	32
accuracy			0.7439	82
macro avg	0.7372	0.7112	0.7177	82
weighted avg	0.7410	0.7439	0.7366	82

The model correctly classified about 74.39% of testing records. The Yes-class precision of 72.00% means that when the model predicted a member as at risk, it was correct about 72% of the time. The Yes-class recall of 56.25% means that the model identified just over half of the actual at-risk members.

For this business problem, recall for the Yes class is especially important because missed at-risk members may not receive retention support before failing to renew. Accuracy alone is not sufficient because the dataset contains more No cases than Yes cases.

10. Confusion Matrix and Business Error Analysis

Table 15: Confusion matrix values

Actual / Predicted	Predicted No	Predicted Yes
Actual No	43	7
Actual Yes	14	18

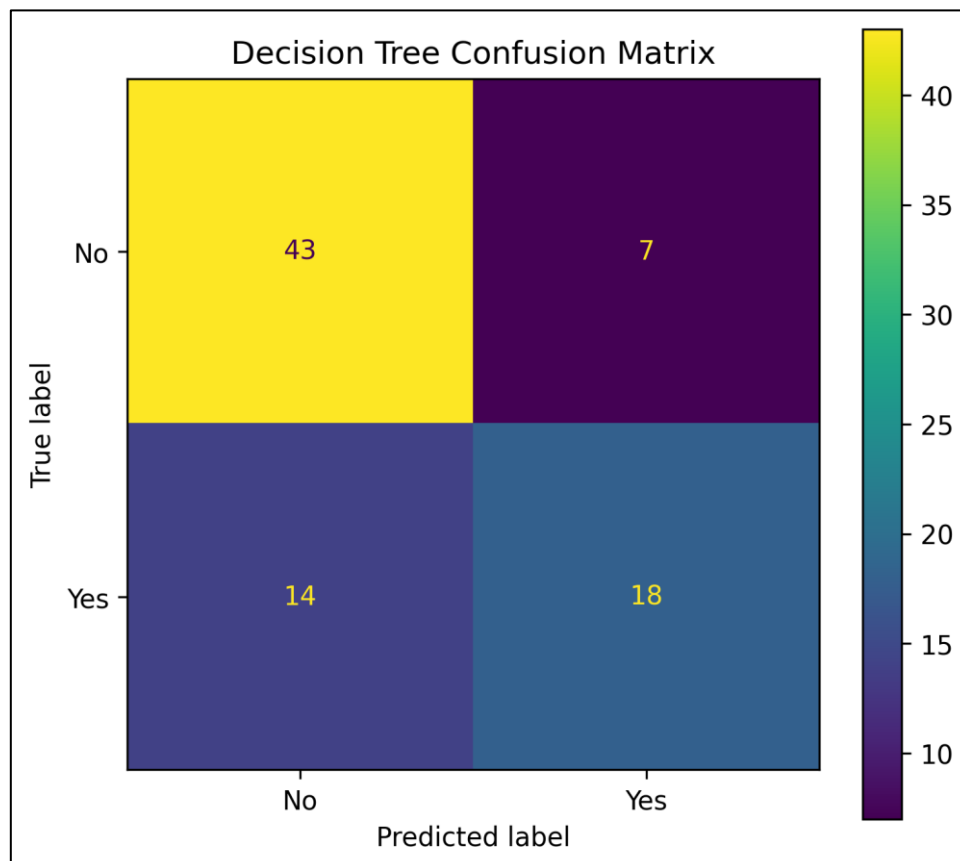


Figure 7: Decision Tree confusion matrix**Table 16: Confusion matrix business meaning**

Item	Value	Business Meaning
True Negative	43	Members correctly predicted as not at risk.
False Positive	7	Members incorrectly predicted as at risk; may receive unnecessary retention contact.
False Negative	14	At-risk members incorrectly predicted as not at risk; may be missed by retention efforts.
True Positive	18	At-risk members correctly identified for possible retention support.

The model produced 18 true positives and 43 true negatives. It also produced 7 false positives and 14 false negatives. In the FitLife context, false negatives are more serious because they represent actual at-risk members who may not receive timely support. False positives still matter because they may create unnecessary staff work or discount costs, but they are generally less harmful than missing a member who may churn.

11. Feature Importance and Business Interpretation

Decision Trees provide feature importance values that indicate which variables contributed most strongly to the model splits. These values help translate model behaviour into practical business insight.

Table 17: Top 10 feature importance values

Feature	Importance	Percentage
Avg_Weekly_Visits	0.145642	14.56%
Classes_Per_Month	0.123850	12.38%
Contract_Type_Monthly	0.120326	12.03%
Satisfaction_Score	0.104301	10.43%
Renewal_Reminder_Sent_Yes	0.097349	9.73%
Membership_Type_Basic	0.088827	8.88%
Payment_Issues_Last_6_Months	0.085698	8.57%
Membership_Type_Premium	0.081710	8.17%
App_Login_Days_Last_30	0.062998	6.30%
Age	0.039688	3.97%

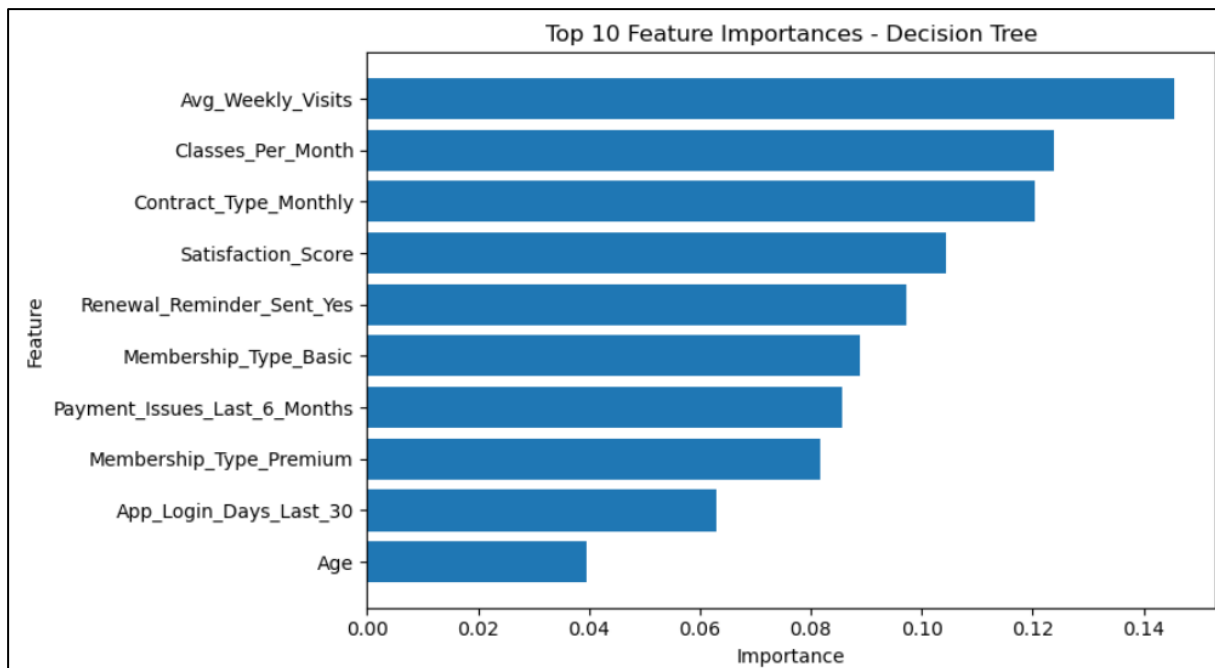


Figure 8: Top 10 feature importances - Decision Tree

The most important features were strongly connected to engagement, contract structure, satisfaction, renewal communication, payment experience, membership type, and age. Average weekly visits and classes per month were the two highest-ranked features, which suggests that member engagement is a central signal of renewal risk.

Some important features, such as `Contract_Type_Monthly`, `Membership_Type_Basic`, and `Membership_Type_Premium`, were created through one-hot encoding. This means that the original categorical variables were converted into separate binary columns before model training.

Feature importance values show which variables were useful for the model's splits, but they do not prove that these variables directly cause non-renewal. Therefore, the results should be interpreted as decision-support insights rather than causal conclusions.

12. Overfitting and Generalization

Table 18: Training and testing accuracy comparison

Evaluation Set	Accuracy	Percentage
Training Set	0.7805	78.05%
Testing Set	0.7439	74.39%
Difference	0.0366	3.66%

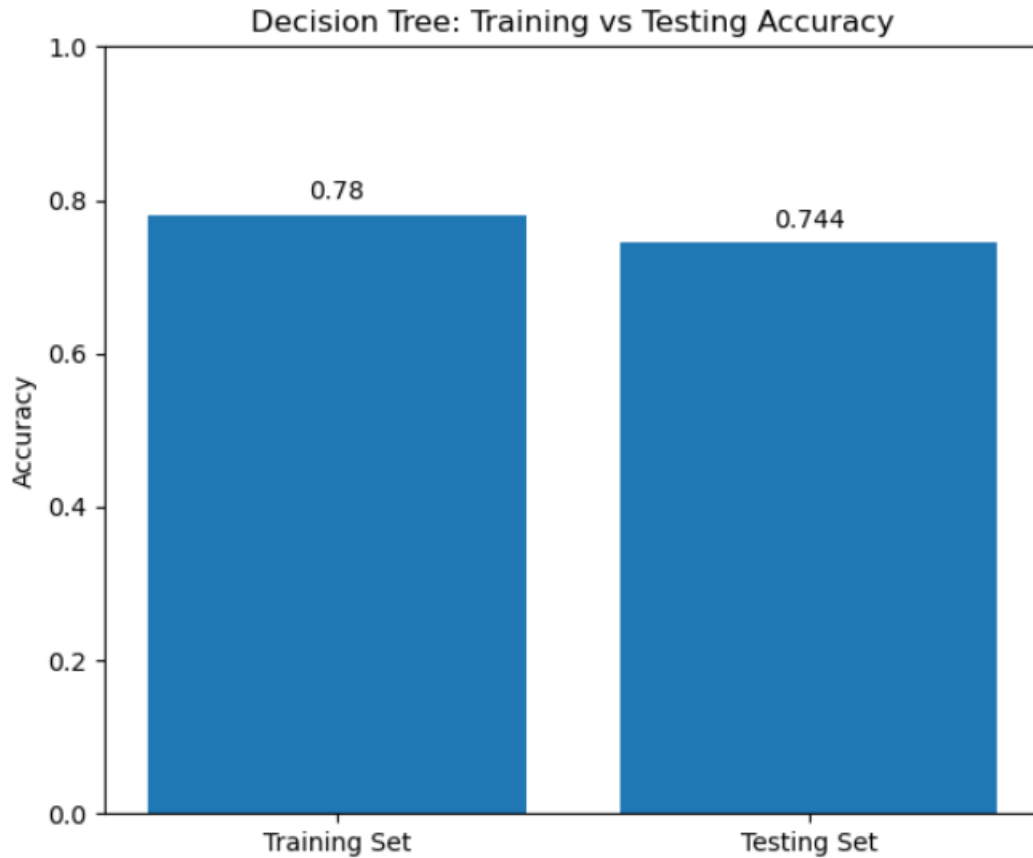


Figure 9: Training accuracy vs testing accuracy

The training accuracy was 78.05%, while the testing accuracy was 74.39%. The difference was approximately 3.66%. This gap is relatively small, so the model does not show strong signs of overfitting. The `max_depth=4` setting appears to support a reasonable balance between model performance and interpretability.

A much larger training-testing gap would suggest that the model memorized patterns in the training data and did not generalize well to unseen members. In this assignment, the tree appears reasonably balanced for a first Decision Tree model.

13. Business Recommendations

The model should be used as a screening tool to support customer retention planning. FitLife can use the predictions and important features to prioritize member outreach, but human review should guide final actions.

Table 19: Business recommendations linked to important features

Signal / Feature	Business Interpretation	Recommended Action
Low average weekly visits	Members may not be receiving enough value from the membership.	Send class suggestions, check-in messages, or goal-setting support.

Low classes per month	Reduced class participation can signal disengagement.	Recommend beginner-friendly or preferred class types.
Monthly contract type	Monthly members can exit more easily than long-term members.	Offer renewal incentives or explain six-month/annual benefits.
Low satisfaction score	Customer experience problems may increase churn risk.	Arrange service recovery call or manager follow-up.
No renewal reminder or weak response	The member may miss renewal timing or ignore communication.	Use timely reminders through email, app, phone, or in-person follow-up.
Payment issues	Billing friction can push members toward cancellation.	Provide billing support before the renewal date.
Low app login activity	Digital disengagement may signal lower relationship strength.	Encourage app use through booking tips, progress tracking, and class alerts.

A practical implementation approach would be to review members predicted as Yes each week, combine the model output with staff knowledge, and assign the most appropriate retention action. The model should not automatically trigger discounts for every predicted at-risk member; instead, the business should match the intervention to the reason for risk.

14. Limitations, Bias, and Responsible AI

The model has useful business value, but it also has limitations. The dataset is synthetic and relatively small, and it may not include all factors that influence renewal decisions. Real members may leave or renew for reasons such as health, family responsibilities, relocation, transportation, income changes, work schedules, competitor offers, or personal preferences.

Table 20: Limitations and mitigation plan

Limitation	Business Risk	Mitigation / Future Improvement
Synthetic and limited dataset	The data may not fully represent real customer behavior.	Validate the model using real historical FitLife renewal data before operational use.
Moderate recall for Yes class	Some at-risk members are missed by the model.	Use the model as a screening tool and monitor recall over time.
Potential class imbalance effect	The No class is larger than the Yes class.	Evaluate precision, recall, F1-score, and confusion matrix, not accuracy alone.
Possible bias by membership type or location	Some member groups may be over- or under-flagged.	Check performance across membership types, distance groups, and demographic categories.
No cost-sensitive threshold tuning	The model may not reflect the relative cost of false negatives and false positives.	Consider recall-focused tuning or cost-sensitive analysis if retention cost data becomes available.

Responsible use requires human judgment, fairness review, and continuous performance monitoring. FitLife should avoid treating model predictions as final truth. A member predicted as not at risk may still need support, and a member predicted as at risk may have a temporary reason for lower engagement.

15. Conclusion

This assignment developed a Decision Tree classification model to predict whether FitLife Studio members are at risk of not renewing their membership. The workflow included business understanding, data inspection, duplicate removal, missing value inspection, feature and target separation, train-test splitting, preprocessing with imputation, model training, model evaluation, feature importance analysis, business interpretation, limitation discussion, and responsible AI reflection.

The final Decision Tree model achieved 74.39% testing accuracy, 72.00% precision for the Yes class, 56.25% recall for the Yes class, and 63.16% F1-score for the Yes class. The confusion matrix showed that false negatives remain a key business concern because missed at-risk members may not receive timely retention support.

Overall, the Decision Tree model provides a useful and interpretable starting point for renewal-risk prediction. The model can help FitLife move from reactive cancellation handling to proactive retention management. However, because the model does not identify every at-risk member, it should support human judgment rather than replace it.

Appendix A: Decision Tree Visualization and Assignment Alignment

The following figure shows the fitted Decision Tree structure used for the final model. Because the model uses one-hot encoded categorical variables, several split variables appear as encoded feature names.

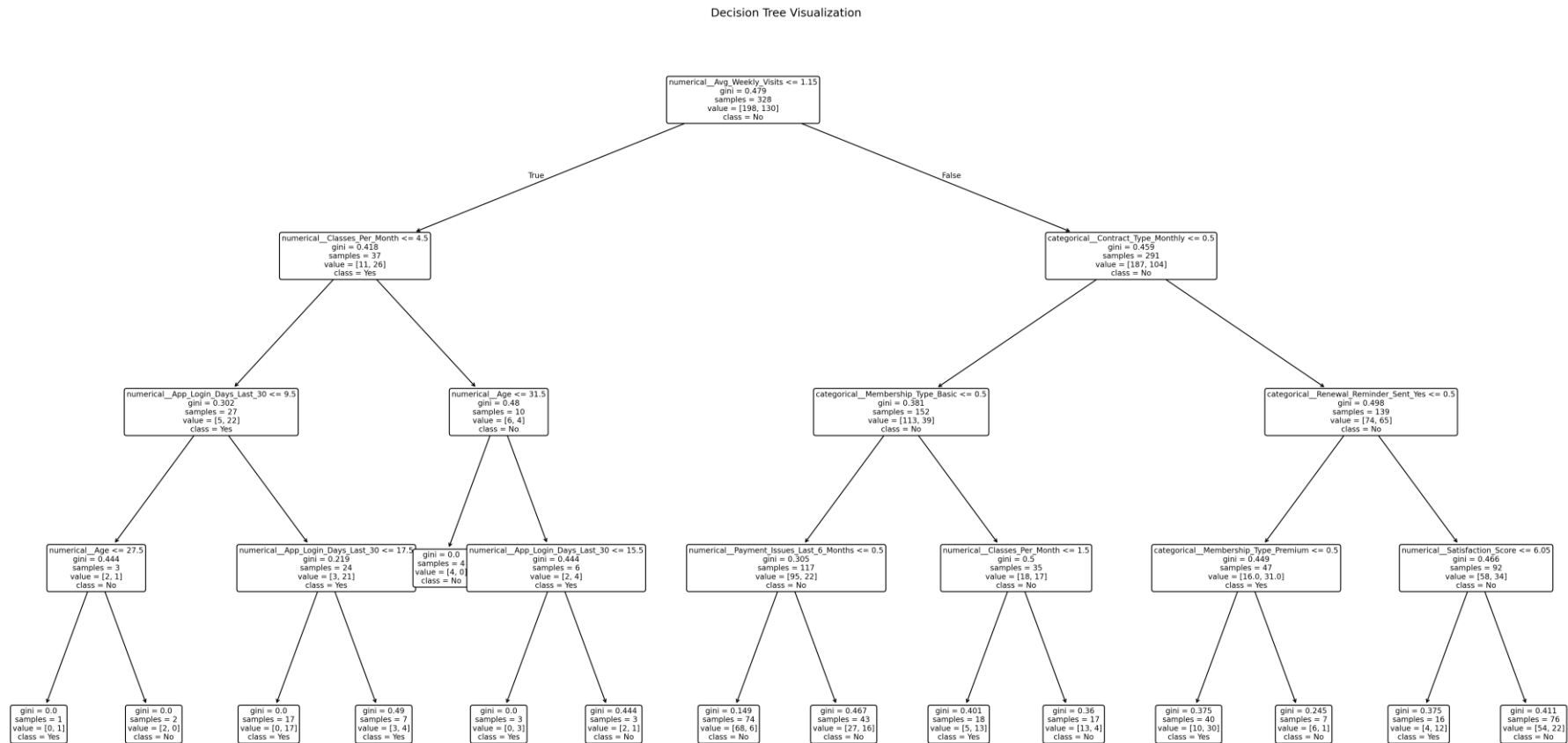


Figure 10: Decision Tree visualization

Table 21: Assignment alignment checklist

Assignment Requirement	Where Addressed in This Report
Understand the business problem	Sections 3 and 13 explain the business context, churn problem, and retention actions.
Prepare and clean the data	Sections 4 to 7 document duplicates, missing values, feature groups, and train-test split.
Build a Decision Tree model	Section 8 explains preprocessing and Decision Tree configuration.
Evaluate model performance	Sections 9 and 10 provide accuracy, precision, recall, F1-score, classification report, and confusion matrix.
Check overfitting	Section 12 compares training accuracy and testing accuracy.
Interpret feature importance	Section 11 explains the top 10 important features and business meaning.
Discuss limitations and human judgment	Section 14 explains limitations, possible bias, and responsible AI use.